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 Studierichting: Informatica
 Oefeningenreeks 2B nr. 6.9

Toon aan dat $5 \operatorname{Bgtan} \frac{1}{7} + 2 \operatorname{Bgtan} \frac{3}{79} = \frac{\pi}{4}$.

$\alpha = \operatorname{Bgtan} \frac{1}{7}$ en $\beta = \operatorname{Bgtan} \frac{3}{79} \Rightarrow \tan \alpha = \frac{1}{7}$ en $\tan \beta = \frac{3}{79}$

We gebruiken: $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

$$\begin{aligned} \tan 2\alpha &= \frac{2 \cdot \frac{1}{7}}{1 - \left(\frac{1}{7}\right)^2} \\ &= \frac{\frac{2}{7}}{1 - \frac{1}{49}} = \frac{\frac{2}{7}}{\frac{48}{49}} \\ &= \frac{7}{24} \end{aligned}$$

Somformule geeft:

$$\tan 3\alpha = \tan(2\alpha + \alpha)$$

$$\begin{aligned} &= \frac{\frac{7}{24} + \frac{1}{7}}{1 - \frac{7}{24} \cdot \frac{1}{7}} \\ &= \frac{\frac{73}{24}}{\frac{168}{24}} = \frac{73}{161} \end{aligned}$$

$$\tan 4\alpha = \tan(3\alpha + \alpha)$$

$$\begin{aligned} &= \frac{\frac{73}{161} + \frac{1}{7}}{1 - \frac{73}{161} \cdot \frac{1}{7}} \\ &= \frac{\frac{672}{1127}}{\frac{1054}{1127}} = \frac{336}{527} \end{aligned}$$

$$\tan 5\alpha = \tan(4\alpha + \alpha)$$

$$\begin{aligned} &= \frac{\frac{336}{527} + \frac{1}{7}}{1 - \frac{336}{527} \cdot \frac{1}{7}} \end{aligned}$$

$$= \frac{2879}{3353}$$

Nu $\tan 2\beta$:

$$\begin{aligned}\tan 2\beta &= \frac{2 \cdot \frac{3}{79}}{1 - \left(\frac{3}{79}\right)^2} \\ &= \frac{\frac{6}{79}}{1 - \frac{9}{6241}} = \frac{\frac{6}{79}}{\frac{6232}{6241}} \\ &= \frac{237}{3116}\end{aligned}$$

We bepalen $\tan(5\alpha + 2\beta)$.

Somformule geeft:

$$\begin{aligned}\tan(5\alpha + 2\beta) &= \frac{\frac{2879}{3353} + \frac{237}{3116}}{1 - \frac{2879}{3353} \cdot \frac{237}{3116}} \\ &= \frac{\frac{9765625}{10447948}}{\frac{9765625}{10447948}} \\ &= 1\end{aligned}$$

Dus

$$\begin{aligned}5\alpha + 2\beta &= \text{Bgtan}(1) = \frac{\pi}{4} \\ \Rightarrow 5 \text{Bgtan} \frac{1}{7} + 2 \text{Bgtan} \frac{3}{79} &= \frac{\pi}{4}\end{aligned}$$

References